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i686-linux-musl

This work is based on musl-cross-make's litecross

Preparing for the build

Setting environment variables

Set the following environment variables

```
KERNEL_ARCH="x86"
TARGET="i686-linux-musl"
DOWNLOADS=${PWD}/downloads
SOURCES=${PWD}/sources
OUTPUT="${PWD}/output/${TARGET}"
HOST=$(echo ${MACHTYPE} | sed "s/-[^-]*/-cross/")
MAKEFLAGS="-j$(nproc)"
PATH="${OUTPUT}/bin:${PATH}"
GCC_CONFIG_FOR_TARGET=""

export PATH OUTPUT KERNEL_ARCH TARGET HOST MAKEFLAGS
```

Download and extract the sources

Set the version of the software that we want to download:

```
binutils_ver=2.45
gcc_ver=15.2.0
musl_ver=1.2.5
gmp_ver=6.3.0
mpc_ver=1.3.1
mpfr_ver=4.2.2
linux_ver=6.16.1
```

Download the source tarballs:

```

mkdir -p ${SOURCES} ${DOWNLOADS}; cd ${DOWNLOADS}
wget -c https://sourceware.org/pub/binutils/releases/binutils-$binutils_ver.
tar.xz
wget -c https://ftp.gnu.org/gnu/gcc/gcc-$gcc_ver/gcc-$gcc_ver.tar.xz
wget -c https://musl.libc.org/releases/musl-$musl_ver.tar.gz
wget -c https://ftp.gnu.org/gnu/gmp/gmp-$gmp_ver.tar.xz
wget -c https://ftp.gnu.org/gnu/mpc/mpc-$mpc_ver.tar.gz
wget -c https://ftp.gnu.org/gnu/mpfr/mpfr-$mpfr_ver.tar.xz
wget -c https://www.kernel.org/pub/linux/kernel/v6.x/linux-$linux_ver.tar.xz

```

Extract the sources:

```
for f in *.tar*; do tar xf $f -C ${SOURCES} ; done
```

Create directory structure

```

mkdir -p ${OUTPUT}/${TARGET}
ln -s . ${OUTPUT}/${TARGET}/usr

```

Linux Headers

Enter in the linux kernel source tree and type:

```

cd ${SOURCES}/linux-$linux_ver
make mrproper
ARCH=${KERNEL_ARCH} make headers
find usr/include -type f ! -name '*.h' -delete
cp -rv usr/include ${OUTPUT}/${TARGET}

```

Binutils

Create a build directory for binutils

```

mkdir -p ${SOURCES}/build-binutils
cd ${SOURCES}/build-binutils

```

Configure Binutils:

```

../binutils-$binutils_ver/configure --enable-gprofng=no \
--disable-separate-code \
--disable-werror \
--target=${TARGET} --prefix= \
--libdir=/lib --disable-multilib \
--with-sysroot=${TARGET} \
--enable-deterministic-archives \
--build=${HOST} --host=${HOST}

```

Build and install the package:

```

make
make install-strip DESTDIR=${OUTPUT}

```

Musl Headers

Create a build directory for musl

```
mkdir -p ${SOURCES}/build-musl
cd ${SOURCES}/build-musl
```

Configure and install the headers

```
../musl-$musl_ver/configure --prefix=/usr \
    --host=$TARGET --disable-nls --disable-werror
make DESTDIR=${OUTPUT}/${TARGET} install-headers
```

Keep the source directory but remove the build directory

```
cd ${SOURCES}
rm -rf build-musl
```

GCC (Core)

Enter in the source directory of GCC and prepare GMP, MPFR and MPC

```
cd gcc-$gcc_ver
ln -sf ../gmp-$gmp_ver gmp
ln -sf ../mpfr-$mpfr_ver mpfr
ln -sf ../mpc-$mpc_ver mpc
```

Create a build directory for gcc

```
mkdir -p ${SOURCES}/build-gcc
cd ${SOURCES}/build-gcc
```

Configure GCC:

```
../gcc-$gcc_ver/configure --target=$(TARGET) \
    --host=$(HOST) --build=$(HOST) \
    --enable-languages=c,c++ \
    --disable-bootstrap \
    --disable-assembly \
    --disable-werror \
    --target=$(TARGET) --prefix= \
    --libdir=/lib --disable-multilib \
    --with-sysroot=$(TARGET) \
    --with-build-sysroot=${OUTPUT}/${TARGET} \
    --enable-tls \
    --disable-libmudflap --disable-libsanitizer \
    --disable-gnu-indirect-function \
    --disable-libmpx \
    --enable-initfini-array \
    --enable-libstdcxx-time=rt \
    $(GCC_CONFIG_FOR_TARGET)
```

Build the core of GCC

```
make all-gcc
make install-strip-gcc DESTDIR=${OUTPUT}
```

Build and install libgcc.a

```
make all-target-libgcc enable_shared=no
make install-strip-target-libgcc DESTDIR=${OUTPUT}
```

Leave the build directory intact. We will use it later

Musl libc

Create a build directory for musl

```
mkdir -p ${SOURCES}/build-musl
cd ${SOURCES}/build-musl
```

Configure, build and install Musl libc

```
../musl-${musl_ver}/configure --prefix=/usr \
    --host=${TARGET} --disable-nls --disable-werror
make
make install DESTDIR=${OUTPUT}/${TARGET}
```

GCC (Final)

Now we will finish building GCC. We will first rebuild **libgcc**, this will provide us a shared library.

Enter in the build directory of GCC and clean the build directory of **libgcc**:

```
cd ${SOURCES}/build-gcc
make -C ${TARGET}/libgcc distclean
```

We can now rebuild the support library of the compiler:

```
make all-target-libgcc
make install-strip-target-libgcc DESTDIR=${OUTPUT}
```

Finish the build of GCC and install it:

```
make
make install-strip DESTDIR=${OUTPUT}
```

We should now have a working cross-compiler!